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An Elderly-Oriented Form Design of Low-Speed New Energy Vehicles Based on Rough Set Theory and Support Vector Regression

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Abstract: With the current trend of social aging, the travel needs of the elderly are increasingly prominent. As a means of urban transportation, low-speed new energy vehicles (NEVs) are widely used among the elderly. Many studies are devoted to exploring the function of cars and the travel modes that meet the needs of older people. However, in addition to product performance, the Kansei needs of users also play a key role in communication between enterprises and users. Therefore, the problem of how to improve car shapes in the initial stage of design to meet the Kansei needs of elderly users remains to be solved. In order to fill this gap, the design of low-speed NEVs are selected as the study objects so as to explore the relationship between the visual perception of elderly users and car design; thus, a design method for the form of elderly-oriented cars is proposed. Firstly, using the research framework of Kansei engineering, factor analysis is used to cluster elderly-oriented Kansei factors. Second, the cars' appearances are deconstructed by morphological analysis, and the key design features affecting elderly-oriented satisfaction are identified by a rough set attribute reduction algorithm. Finally, support vector regression is used to establish a mapping model of elderly-oriented Kansei factors and the key design features to predict the elderly-oriented form design of optimal low-speed NEVs. The research results show that selecting "Hub6", "Headlight9", "Car side view2", "Rearview mirror9", and "Front door10" in the form deconstruction table for low-speed NEVs can elicit optimal emotions in elderly users. The research results enable enterprises to more effectively understand the emotional cognition of elderly users related to the form of low-speed NEVs and improve the purchase desire and satisfaction of elderly users, providing references and guidance for the elderly-oriented design and development of intelligent transportation tools.

Keywords: elderly-oriented design; Kansei engineering; rough set theory; support vector regression; low-speed electric vehicle

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1. Introduction

Under the aggravation of global environmental problems and the increasing urgency of the energy crisis [1], many countries, including Japan and the United Kingdom, have declared projects that will suspend the sale of internal combustion engine automobiles within the next 20 years. In addition, countries around the world have successively introduced a variety of policies and measures to support the development of NEVs, aiming to realize the green transformation of the automobile industry through technological innovation and industrial upgrading [2]. In such an environment, NEVs, with their significant advantages in energy savings, environmental protection, and low carbon emissions, have become very popular and are rapidly growing in the global market [3]. New energy power systems are not only a technological change but also redefine the creative space and boundaries of automotive form design. At the same time, the era of Kansei consumption has promoted improvements in the aesthetics and emotional needs considered in automobile modeling design. This trend is significantly reflected in the form design of NEVs [4]. The shape design of an automobile plays an indispensable role in conveying a brand's visual identity

and promoting design innovation and brand differentiation, which has a crucial impact on consumers' purchase decisions [5].

At the same time, with the increasingly prominent problem of population aging in countries around the world, governments need to find strategies to solve the travel problems of the elderly [6,7]. In the current research literature, based on ANNCF-E, Jian et al. [8] quantify the impact of elderly drivers on traffic safety by introducing alternative safety indicators. Lu et al. [9] discuss whether elderly users need wider parking spaces and propose a new parking concept scheme to help elderly users park more smoothly. Lajunen et al. [10] conduct a study on the attitude of elderly drivers toward the four main levels of autonomous driving technology, and the study shows that elderly users are more inclined to use SAE2 cars. In the development of autonomous vehicles, elderly drivers' attitudes toward automation should be fully considered to develop the full potential of automation. Fatima et al. [11] conduct a literature review on sustainable travel for the elderly and point out that future studies should consider the dominant role of private transportation in promoting travel for the elderly and its impact on cities with aging populations. Against this background, the key point of automobile research for elderly users is mainly the realization of the functions and prospects of future travel. However, the problem of how to improve car shapes in the initial stage of design to meet the emotional needs of elderly users remains to be solved. The popularity of NEVs, especially emergency low-speed NEVs, has greatly improved travel for the elderly [12], meeting their mobility needs and providing new impetus for the sustainable development of smart cities [13]. As the global population is increasingly growing, the United Nations emphasizes that countries should fully consider the influence and specific requirements of population aging when formulating policies [14], which has made the issue of an aging global population the focus of the 21st century's global community [15,16]. Nowadays, elderly users of NEVs have become a key force in the consumer market. As for low-speed NEVs, they have become a major challenge in the fields of design, research, and development. The implementation of elderly-oriented design is not only related to the safe travel of the elderly and improvement in their travel autonomy [17,18], but it also plays a crucial part in promoting the expansion and innovation of the NEV market, which is consistent with the grand objective of sustainable social development. Because of changes in body functions and perception [19], elderly NEV users have put forward higher standards for the safety, comport, and ease of travel tools. Therefore, in the initial stage of product design, enterprises must take the physical functions and emotional needs of elderly customers into account, thus creating and providing more convenient and intimate travel solutions [20].

In Kansei research on such users, KE is defined as an emerging technology with a core focus on the emotional needs of users [21], which combines multidisciplinary knowledge, such as engineering technology, psychology, and ergonomics [22], and is committed to converting users' perceived experiences and intentions into specific product design parameters. KE was first applied to product design [23], and it was then realized in the automotive industry [24]. After half a century of evolution, KE has not only made remarkable achievements in the field of automotive design [25,26] but has also been widely applied in various product designs [27,28], clothing designs [29], and service designs [30]. When designing products for elderly users, we must recognize that their emotional needs are often characterized by uncertainty, variability, and ambiguity; their capture and resolution often rely on nonlinear data processing methods. Based on this, the KE research framework is usually divided into the following three stages: (1) summaries and cluster analyses of users' Kansei demand factors; (2) identification of key design features of samples; and (3) building of a mapping model between key design features and users' Kansei needs. On the one hand, in the second stage, it is necessary to use comprehensive methods to assist in making design decisions and in determining the most influential design characteristics, in addition to product forms, which can improve the computational efficiency of subsequent studies. Traditional design support methods are used, for instance, the analytic hierarchy process [31] and fuzzy comprehensive evaluation [32]. When assisting design decisions, the

weights and interactions of many factors are taken into consideration. On the other hand, in the third stage, the key task of KE is to set up a correlation model between the users' emotional needs and the product design features so as to predict the design scheme that will resonate with the users. Traditional methods, such as the kano model [33] and quality function deployment [34], rely on expert ratings and users' surveys to obtain key data, using mathematical models and charts to conduct an analysis of the correlation between the characteristics of the design and the emotional needs of users. The above traditional decision-making methods often face strong subjectivity and a limited ability to process data, which makes their effective use difficult due to the limitations of complex or fuzzy situations. Therefore, artificial intelligence technologies such as machine learning and deep learning have been widely used to establish complex databases and develop advanced reasoning mechanism to provide effective decision support for the product design process [35]. Artificial intelligence algorithms can learn patterns and features from large amounts of data through algorithm models, which not only reduces human bias but also provides greater accuracy and an ability to process large-scale data sets [36]. RST is a tool based on mathematical theory put forward by Polish scholar Pawlak, in 1982, which allows users to deal with imprecise, uncertain, or ambiguous knowledge and data [37]. When it is applied to KE, the attribute reduction algorithm of RST does not require any prior knowledge, and it can reveal the most important attributes through the data themselves and retain the most critical attributes in the decision-making process [38]. SVR is an advanced machine learning method based on systems learning theory [39], which is specially used to solve the problems of classification and regression [40]. By introducing kernel functions, it can process linear indivisible data, complete regression analysis, identify and build complex model relationships, and it is very suitable for predicting complex mapping patterns that may exist between the characteristic of the design and the emotional needs of users [41].

Therefore, a combination method for the elderly-oriented design of the product's form combing RST with SVR on the basis of the KE framework is put forward in this paper. The low-speed NEV is taken as an example to verify the feasibility of this method. Within the framework of KE, the elderly-oriented Kansei prediction of a low-speed NEV is carried out. RST can identify the design features of key low-speed NEVs, and SVR constructs the mapping relationship between the elderly users' Kansei needs and the low-speed NEV. In accordance with a survey of the literature, there are no studies on product shapes in elderly-oriented design from a KE perspective. The primary objective of this study was to utilize RST and SVR in the elderly-oriented design of low-speed NEVs. The representative emotions of elderly users are identified through in-depth interviews and Kansei questionnaire research. Building upon this foundation, the optimal combination of elderly-oriented design features for low-speed NEVs is established using artificial intelligence technology to effectively cater to the emotional needs of elderly users. The elderly-oriented design of low-speed NEVs not only helps to meet the travel needs of elderly groups, but also expands the NEV consumer base, which is beneficial for increasing the production and sales of NEVs and reducing the environmental pollution from fuel-based vehicles. Thus, the ultimate goal of the green and sustainable development of society will be achieved. From the research results, the best combination of elderly-oriented product forms for low-speed NEVs with the highest average Kansei evaluation by elderly users is obtained, which helps enterprises expand their elderly consumer groups, improves users' satisfaction, and saves time and costs in elderly-oriented product development.

The structure of this research is organized as follows: Section 2 discusses the methodology employed; Section 3 introduces the research framework for designing products tailored to the elderly; Section 4 presents an analysis and discussion of the study's findings; finally, in Section 5, the author summarizes the conclusions drawn from the research and suggests potential avenues for future research.

2. Relative Methods

2.1. Kansei Engineering

KE is a method of product development that originated at Hiroshima University in Japan. It focuses on capturing and transforming consumers' emotions and feelings in product design. The application of KE is particularly important in the automotive industry, and it can help manufacturers design cars that inspire specific feelings in consumers and meet their emotional needs. This engineering approach has influenced the styling design of automobiles since the 1980s [24], and it has gradually been widely adopted in car design, especially in Japan and then subsequently worldwide. Initially, the application of KE focused on the color and texture selections of car interiors and small parts to create a comfortable driving environment in line with consumers' emotional expectations [25,42]. In the 21st century, with the developments in technology and the diversification of consumers' preferences, KE has started to be used in more depth to design car models. Among them, in the design of the front faces of cars, "face theory" is used, which allows for the expression of the "character" of a car by adjusting the layout of the headlights and grille [43]. Modern KE is not limited to vision and touch but also extends to the design experience of car users, including sound engineering and the interactive ways of cars [18]. Famous automobile manufacturers around the world, such as Toyota, Mazda, Honda, BMW, and Audi, use the KE principle, to different degrees, thus guiding the design of cars that receive high praise. In existing academic research, Lai et al. [44] put forward a hybrid Apriori+ structural equation model system, which finds that some traditional design features of cars may have opposite effects for NEVs, and they also propose the development of a design assistance system. Lin et al. [45] propose an NLP-MDCEV model to analyze the Kansei demand and behavior data of users for the purpose of solving priority problems related to design tasks. The application of KE in cars' styling design marks a shift from a single functional orientation to a more comprehensive emotion-oriented design. In order to evaluate car styling, consumer preference assessments [46] have also been applied to existing automobile research. Nevertheless, the existing literature shows that KE tends to be used to focus on young consumer groups in automobile design research and most of the research objects are traditional fuel-based vehicles and high-speed NEVs. Meanwhile, research on the form design of cars that adapts to the needs of elderly users is relatively insufficient, and it needs to be further explored. In existing studies on elderly-oriented design, Li et al. [47] use a KE framework to cluster the emotional needs of elderly users with an intelligent blood glucose detector. Wang et al. [48] establish a mapping model that matches between an internal interface form of elderly-friendly residential buildings and the emotional needs of elderly users using KE words. Chen et al. [18] use a KE semantic vocabulary to establish a mapping relationship between the emotional needs of elderly users and an L3+autonomous vehicle HMI. The above literature shows that knowledge exchange that uses KE words can meet the needs of elderly users and establish a connection between the emotional needs of elderly users and a product. Elderly users are more focused on emotional experiences, so an in-depth understanding of their emotional needs can help designers create products that match their feelings and preferences. Therefore, based on a KE research framework, low-speed new energy vehicles, which are the main means of transportation for the elderly, are taken as a case study, which not only promotes the care and respect of elderly groups in society but also promotes the development of the NEV industry.

2.2. Rough Set Theory

RST is a mathematical theory-based tool put forward by Polish scientist Pawlak, in 1982, for the handling of imprecise [49], uncertain, or ambiguous knowledge and data [50]. Through technology related to knowledge reduction, solutions to problems or classification rules can be deduced by RST; RST can also effectively handle the uncertainty and fuzzy data of Kansei needs [51]. It plays a core role in the study of interactive relationships between the characteristics of products and the emotional needs of consumers in the process of product design, and it can reveal the hidden rules and knowledge in the data [52]. In

existing studies, RST has been widely used in energy loss prediction [53], financial risk prediction [54], and other fields, with the main purpose of reducing the dimensions of influencing factors and improving the efficiency of subsequent algorithms. When applied to KE, the attribute reduction algorithm of RST can handle data with all types of linear or nonlinear characteristics, deal with nonlinear perceptions by human beings, and efficiently mine and simplify key design features and optimize data sets. Tan et al. [55] use RST to extract design knowledge used to support decision making in washing machine models. Wang et al. [56] take the visual design of cars as an example, proposing an association rule mining method combining RST and the continuous Carnot model to extract the prominent relationship between users' emotional needs and products' morphological features. Guo et al. [57] apply RST to solve problems related to KE evaluations for user groups and collective preference information. Wang et al. [58] take a tablet computer as an example; RST is applied to determine the optimal attractive investment portfolio. Liu et al. [59] combine the Kano model and RST to determine users' needs so as to promote quality of service by Chinese academic libraries.

In the process of designing elderly-oriented product models, designers often face the challenge related to the fuzzy perceptions of elderly users and the changeable characteristics of products. The RST's attribute reduction algorithm can explore in depth the internal relationships and importance of design feature parameters, thus effectively locking in the key design characteristics that are crucial to elderly-oriented satisfaction. Therefore, this method was adopted in this study to remove redundant design features, simplify design parameters, and retain key design features that have a significant influence on elderly-oriented satisfaction concerning low-speed NEV forms, thereby improving the efficiency of subsequent calculations.

In RST's attribute reduction algorithm, a weight of 0 for the conditional attribute indicator indicates that the attribute either does not contribute to the reduction process or is unable to differentiate decision attributes [60]. Conditional attributes with a weight of 0 or those at the lowest weight level can be eliminated from the decision table based on reduction principles. This simplification streamlines the decision table, reducing the computational complexity, and improving the algorithm's efficiency and accuracy. The following section offers a brief explanation of RST's principle and attribute reduction algorithm (Figure 1).

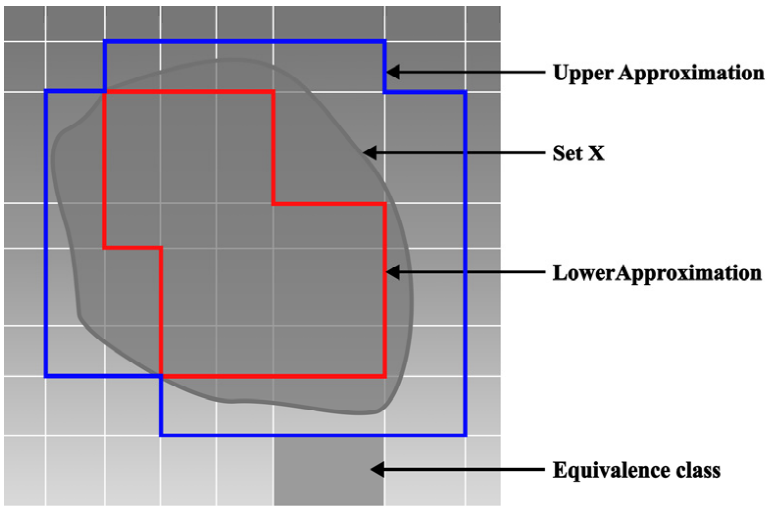


Figure 1. Fundamental concepts in RST.

Definition 1. *It is assumed that* $L = (U, A, V, f)$ *is an information system, also known as a knowledge system, in which* U *represents the nonspatial domain of all assessment records, which is also known as a discourse domain;* A *is a nonempty finite set of attributes,* $A = C \cup D$,

$C \cap D \neq \varnothing$. C is the conditional attribute set, namely, the design feature. D is the decision attribute set, namely, elderly-oriented satisfaction; and V represents the attribute range; f is the relation set of U and A, which is also called the information function set. If $D \neq \varnothing$, the information system L is called the data table; otherwise, the decision information system is referred to as the decision table or is called the decision table for short.

Definition 2. *Let R be the equivalence relation over U, see Formula (1), with the decision attribute and the other equivalence type, namely, $U/IND(D)$. Without considering specific conditional attributes or fields or fields, the equivalence classes are divided on the basis of the original feature set, as follows: $U/Ind(C - \{c_e\})$.*

$$IND(R) = \{(x, y) \in U \times U | \forall a \in A, f(x, a) = f(y, a)\} \quad (1)$$

Definition 3. *In RST feature selection and data analysis, two key concepts, borrowed from RST, are used to determine the level of knowledge approximation, which are the lower approximate set and upper approximate set. $\underline{R}(X)$ is the lower approximate set of R, as shown in Formula (2); $\overline{R}(X)$ is the upper approximation set of R, which is shown in Formula (3), and it is also called the positive field of X. It is expressed as $POS_R(X)$. When $\underline{R}(X) = \overline{R}(X)$, X is an exact set of R; when $\underline{R}(X) \neq \overline{R}(X)$, X is a R rough set. Therefore, the positive field of the conditional attribute set is represented by $pos_c(D)$ and $pos_{c-|c_e|}(D)$.*

$$\underline{R}(X) = \{x \in U | IND(R) \subseteq X\} \quad (2)$$

$$\overline{R}(X) = \{x \in U | IND(R) \subseteq X\} \quad (3)$$

Definition 4. *The decision attribute D's degree of dependence on the condition attribute C is $r_c(D)$. The decision attribute D's degree of dependence on the knowledge attribute $C - \{c_e\}$ is $r_{c-|c_e|}(D)$, and the condition attribute's importance to the decision attribute is $\sigma(c_e)$, as shown in Formula (4).*

$$\sigma(c_e) = r_c(D) - r_{c-|c_e|}(D) \quad (4)$$

Definition 5. *The weight coefficient of the condition attribute indicator is W_e, as shown in Formula (5). In this study, the conditional attributes with a weight of 0 or ranked last rank were deleted.*

$$W_e = \frac{\sigma(c_e)}{\sum_{\lambda=1}^{n} \sigma(c_\lambda)} \quad (5)$$

2.3. Support Vector Regression

The SVR algorithm is an advanced machine learning method based on statistical learning theory [35], which is specially used to solve regression problems [40]. Its main advantage is that it can find the global optimal solution in limited sample data, avoiding the risk of model overfitting and ensuring the generalization ability of the model. SVR provides an effective way to construct a nonlinear relationship mapping model between the design features of products and the emotional needs of users. Kang et al. [61] use the SVR algorithm to develop a mapping model between young consumers and the shape design features of products, thus predicting the body shape of future hybrid electric vehicles. Yang et al. [27] use the SVR algorithm to set up a prediction model for the shape design of an ear thermometer, scoring and filtering the graphs generated by the diffusion model (DM) so as to obtain images with the highest scores to assist designers in developing detailed designs. Yuan et al. [62] use the SVR model to predict the optimal combination of nursing bed design features that meet users' Kansei needs. Yang et al. [63] use SVR to set up a correlation model between the preferences of users and design factors, thus effectively predicting users' preferences. In summary, SVR can be employed to establish a mapping model between users' Kansei intentions and the design characteristics of a product's shape. Therefore, SVR was used to build the design features of a low-speed NEV form and the

mapping model of elderly users' Kansei intention, thus predicting the future body shapes of low-speed NEVs that conform to the Kansei intention of elderly users.

Data prediction was achieved through SVR by means of fitting the sample data within interval zones. The aim was to find a function and construct interval zones on both sides of the function so that the interval zones were as far apart as possible and the loss was as small as possible. In SVR regression analysis, R-square (R^2) is usually used to measure the model's degree of fit with the data. If the value of R^2 is closer to 1, the fitting degree of the model will be better. Secondly, in SVR prediction analysis, mean squared error (MSE) and mean bias error (MBE) mainly measure the difference between the predicted value and the actual value of the model; the smaller the MBE and MSE values, the higher the prediction accuracy of the model.

The basic principle (Figure 2) and calculation method of SVR are briefly introduced in the following contents:

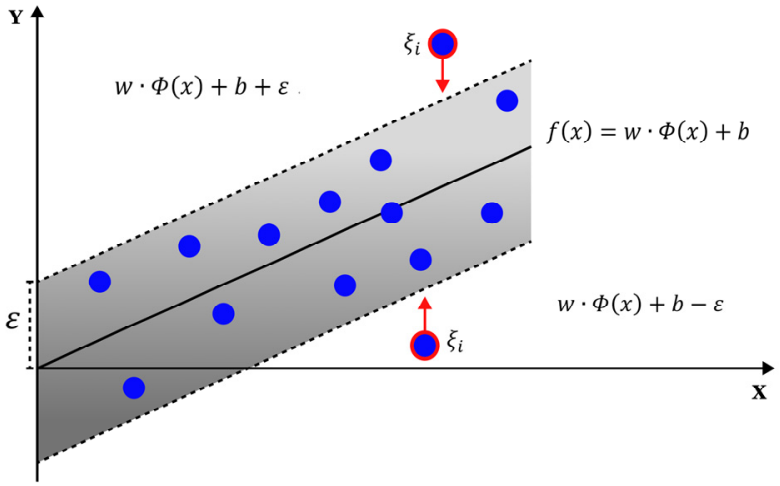


Figure 2. Basic principles of the SVR.

Definition 6 . *For the given data set* (x_i, y_i), $i = 1...n$, x_i *is the input feature vector,* y_i *is the corresponding target value, and n is the total number of data sets. The regression function is constructed as shown in Formula (6), where* $f(x)$ *is the predicted value of the function, w is the weight vector,* $\Phi(x)$ *is the nonlinear mapping function, and b is a bias vector.*

$$f(x) = w \cdot \Phi(x) + b \tag{6}$$

Definition 7. *In order to obtain the vectors w and b, the objective function is minimized as shown in Formulas (7) and (8), where C represents the penalty coefficient, n stands for the number of samples in the database,* x_i *denotes the eigenvector of the i-th sampleis,* $f(x_i)$ *represents the predicted value of the i-th sample,* y_i *is the actual value of the i-th training sample,* $L_\varepsilon[f(x_i), y_i]$ *represents the insensitive loss function, and ε is the maximum allowable error in the regression analysis.*

$$min \frac{1}{2} \parallel w \parallel^2 + C \sum_{i=1}^{n} L_\varepsilon[f(x_i), y_i] \tag{7}$$

$$L_\varepsilon[f(x_i), y_i] = \begin{cases} 0, |f(x_i) - y_i| \le \varepsilon \\ |f(x_i) - y_i| - \varepsilon, |f(x_i) - y_i| > \varepsilon \end{cases} \tag{8}$$

Definition 8. *In consideration of the fitting error, the positive definite relaxation factors xi and xi* of sample No. i are introduced, and the expression of the regression problem is shown in Formula (9), as follows:*

$$min \frac{1}{2} \parallel w \parallel^2 + C \sum_{i=1}^{n} \left[\xi_i + \zeta_i^* \right]$$

$$s.t.\begin{cases} f(x_i) - y_i \le \varepsilon + \xi_i \\ y_i - f(x_i) \le \varepsilon + \xi_i \\ \xi_i \ge 0 \\ \xi_i^* \ge 0 \end{cases}, i = 1, 2, \cdots n \tag{9}$$

Definition 9. *The Lagrange function is introduced to transform the constrained optimization problem into a dual optimization problem, as shown in Formula (10),* where α_i *and* α_i^* *are Lagrange factors, and* $K(x_i, x_j)$ *is the kernel function.*

$$max\left[-\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left(\alpha_i - \alpha_i^*\right)\left(\alpha_j - \alpha_j^*\right)K(x_i, x_j) - \varepsilon\sum_{i=1}^{n}\left(\alpha_i + \alpha_i^*\right) + \sum_{i=1}^{n} y_i\left(\alpha_i - \alpha_i^*\right)\right]$$

$$s.t.\begin{cases} \sum_{i=1}^{n}\left(\alpha_i - \alpha_i^*\right) = 0 \\ 0 \le \alpha_i \le C \\ 0 \le \alpha_i^* \le C \end{cases} \tag{10}$$

Definition 10. *The SVR problem is transformed into a programming problem of the convex function, resulting in a nonlinear regression function, as shown in Formula (11).* When $\gamma = \frac{1}{2\sigma^2}$, *the radial basis kernel function is represented as shown in Formula (12).*

$$f(x) = \sum_{i=1}^{n}\left(\alpha_i - \alpha_i^*\right)K(x_i, x) + b \tag{11}$$

$$K(x_i, x) = \exp\left(-\gamma \cdot \parallel x_i - x \parallel^2\right) \tag{12}$$

Definition 11 .*Kernel function can replace the original inner product and simplify the calculation process, which has a higher fitting. In SVR, the radial basis kernel function is widely used in similarity measurement and feature space mapping to deal with nonlinear problems. In this study, the radial basis kernel function is used. After the training, the determination coefficient* R-squared (R^2) *is adopted to verify the fitting effect of the SVR model. The mean bias error (MBE) is used to measure the predictive performance of the model. The calculation formula is shown below, where y represents the average of the sample's actual value.*

$$\mathrm{R}^2 = 1 - \frac{\sum_{\mathrm{i}=1}^{\mathrm{n}}\left(\mathrm{y_i} - f(x_i)\right)^2}{\sum_{\mathrm{i}=1}^{\mathrm{n}}\left(\mathrm{y_i} - \overline{\mathrm{y}}\right)^2} \tag{13}$$

$$\mathrm{MBE} = \frac{\sum_{\mathrm{i}=1}^{\mathrm{n}}\left(\mathrm{y_i} - f(x_i)\right)}{n} \tag{14}$$

$$\mathrm{MSE} = \frac{\sum_{i=1}^{n}\left(y_i - f(x_i)\right)^2}{n} \tag{15}$$

3. Proposed Research Framework

The objective of this study is to propose a combination method of a product shape's elderly-oriented design with the aid of artificial intelligence technology. Qualitative and quantitative modeling methods were adopted in this study, and the research objects were low-speed NEVs, which is a transportation tool for the elderly. In this paper, a 45-degree squint for low-speed NEVs was selected as the sample angle. Firstly, the Kansei word aggregation process and simplification of the design type of this study are described, and their comprehensive effectiveness is evaluated by combining qualitative and quantitative means. The specific research steps are as follows:

Firstly, FA is utilized for dimensionality reduction among Kansei words to cluster factors specific to the elderly. Following this, the design attributes of a low-speed shape were analyzed through a morphological analysis. Subsequently, the RST attribute reduction algorithm was applied to calculate the weights of the design attributes and identify the

key characteristics that significantly impact elderly satisfaction. Next, SVR was employed to create a mapping model for various elderly-oriented Kansei words. Finally, the design practice was completed according to the experimental results, and the subjects evaluated the design practice to verify the feasibility of the combined method for the elderly-oriented design of products. The specific framework of this research is shown in Figure 3.

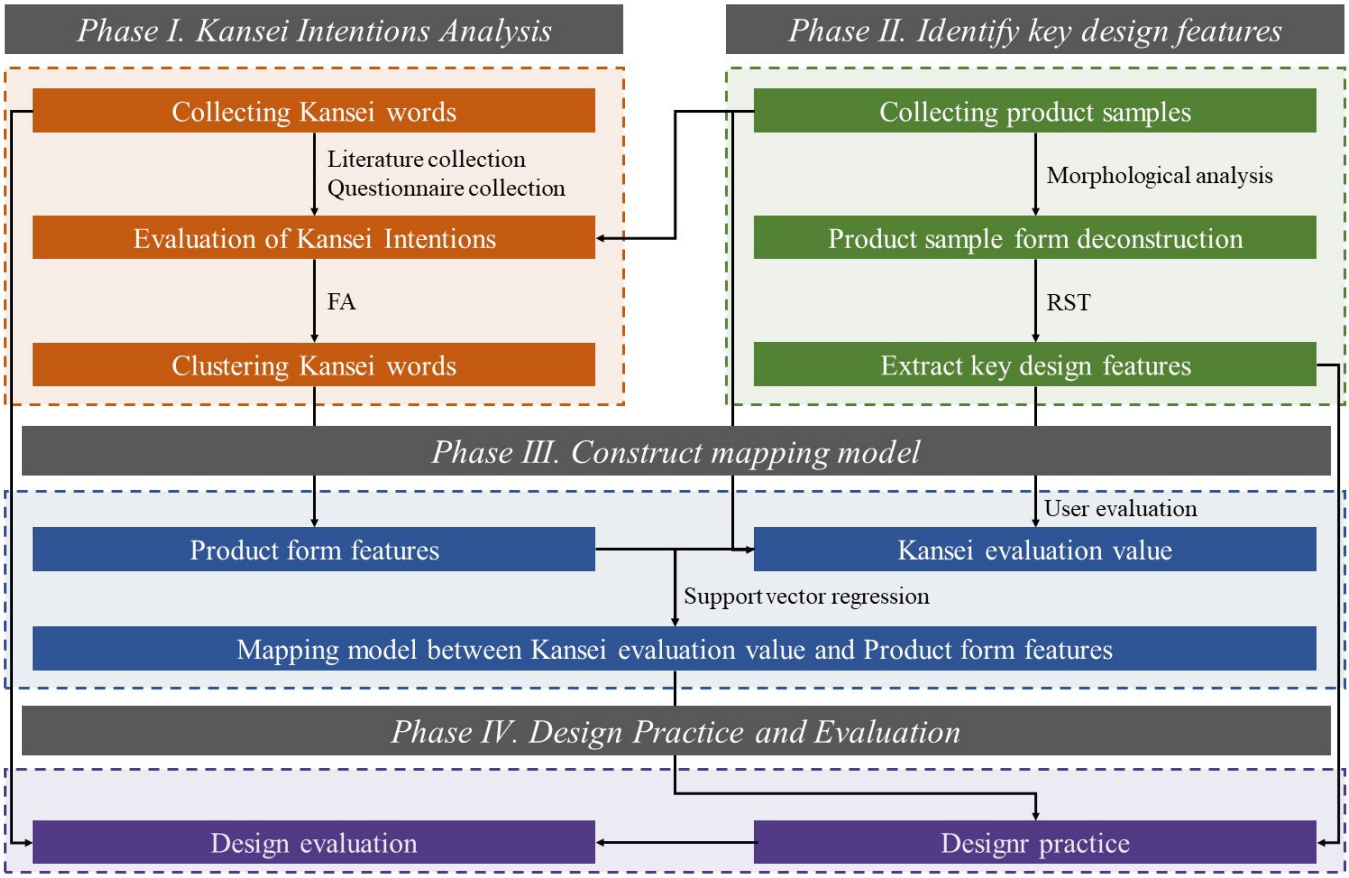


Figure 3. The research framework for elderly-oriented product form design.

3.1. Data Set Sample Determination

In this study, a 45-degree squint for low-speed NEVs was selected as the sample angle. In this thesis, a sample comprising 140 low-speed new energy vehicles in line with the scope of the study was collected through automotive websites. After discussion among an expert group, a total of 100 low-speed NEV samples were screened out by eliminating blur, occlusion, and highly similar samples of the same brand shape, and a low-speed NEV database was created (Figure 4). Considering that different colors for low-speed NEVs will affect the emotional evaluation by the elderly of the car shape, we imported the samples into Photoshop for processing, as shown in Figure 5.



Figure 4. 100 representative low-speed NEVs.

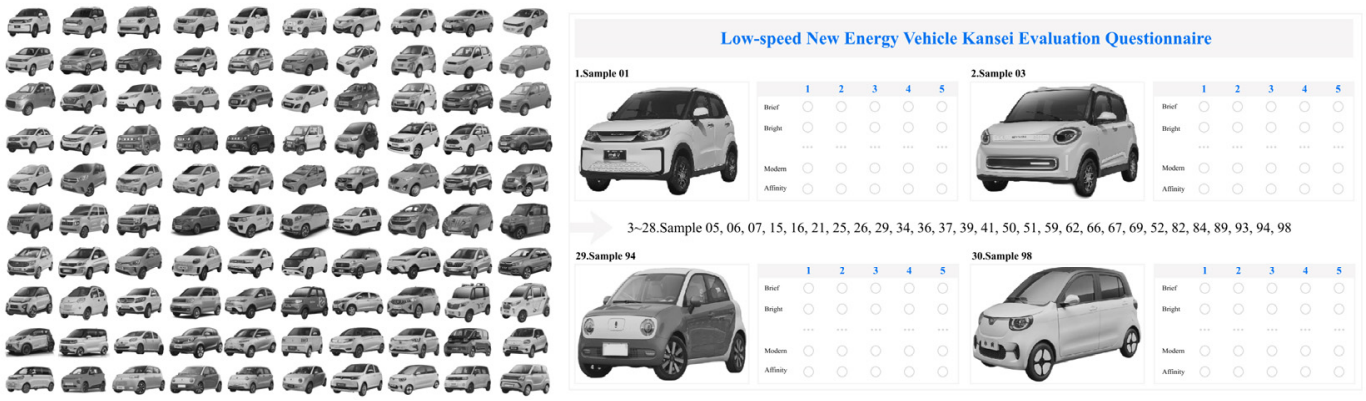


Figure 5. 30 representative low-speed NEVs for the Kansei evaluation.

3.2. Factor Analysis Clustering of Elderly-Oriented Kansei Factors

When the elderly-oriented Kansei words related to low-speed NEVs are collected, they should easily express the target user group's emotional needs and preferences concerning automobile modeling, focusing on emotional words that describe the design style, feelings, and emotional responses related to low-speed NEVs' forms. First, the expert group evaluated and discussed the list of Kansei words related to the low-speed NEV samples in the database. At the same time, 12 Kansei words were initially selected by consulting the relevant literature and automobile websites. In consideration of the psychological and

physical characteristics of the elderly, it is difficult to maintain a high level of concentration. Therefore, the expert group first simplified the 12 Kansei words into 9 Kansei words after discussion, as shown in Table 1. Later, the expert group screened 100 low-speed NEV samples in the sample bank according to the similarity and brand differences. Thirty typical samples (1, 3, 5, 6, 7, 15, 16, 21, 25, 26, 29, 34, 36, 37, 39, 41, 50, 51, 59, 62, 66, 67, 69, 52, 82, 84, 89, 93, 94, and 98) were selected for elderly users to evaluate in terms of Kansei intention, thus reducing the burden on elderly subjects. This study included 200 participants, of which 24 were car designers and 176 elderly users. They were invited to participate in the assessments of their Kansei intentions concerning low-speed new energy vehicles. The 176 elderly users in the study were evenly distributed by gender, with half men and half women. They had 5 years or more than 5 years' driving experience, and with ages between 40 and 65. They were from Nanjing. The participants' specific task was to conduct evaluations of 100 types of low-speed NEV, assessing their perceptions and emotional responses to the nine types of Kansei words. To measure emotional responses, a 5-step Likert scale was used as the primary tool in this study. Participants were asked to objectively record their emotional responses to different car samples and different Kansei words, thus assessing their emotional preferences. In the course of the study, the car designers conducted evaluations through online questionnaires, while the elderly users participated in the study by using offline recording methods at the community activity center. This separate approach can better adapt to the participation needs and circumstances of different groups.

Table 1. Nine representative elderly-oriented Kansei words.

Brief	Bright	Harmonious	Stable	Affinity	Clear	Distinctiveness	Modern	Environmentalism

In this study, FA was used to reduce the dimensions and cluster the Kansei words; the data in Table 2 were imported into the Statistical Package for factor analysis. Firstly, KMO and Bartlett sphericity tests on factor roots were needed to determine whether the data were suitable for the analysis of factors. The results are shown in Table 3, with a KMO value of 0.861. The Bartlett approximation was 189.135, the degree of freedom was 36, and the significance was $0 < 0.05$. The KMO and Bartlett sphericity test results showed that the data were suitable for factor analysis and had significant differences. Secondly, for the total variance interpretation matrix data of users' Kansei intention measurement scales (Table 4), the cumulative contribution rate of the first three indicators was 83.637, which is close to 85%, proving that the nine words could be reduced to three main factors.

The users' Kansei image measurement table underwent orthogonal rotation using the Caesar normalization maximum variance method to adjust Kansei factors. The resulting component matrix is displayed in Table 5, with factors below an absolute value of 0.5 left as blank spaces to minimize visual distractions.

Principal component analysis was applied to obtain the component score coefficient matrix, which is shown in Table 6. In the low-speed NEV Kansei words' factor analysis, we clustered a total of three elderly-oriented Kansei factors. The naming of the factors and the selection of indicators were conducted on the basis of a comprehensive consideration and understanding of the emotional requirements and emotional resonances of the elderly. The first factor was composed of the following five indexes: bright, harmonious, stable, affinity, and differentiation degree, and it was named as the welcome factor. It conveys the warm, friendly, and comfortable sense of welcome that a car's outward appearance elicits in the elderly users, in other words, the exterior appearance of the car and the feeling of a friendly design. The second factor consisted of the following two indicators: brief and clear, and it was named the clarity factor. This factor highlights a barrier-free feeling and the comprehensibility of a car's exterior design, reflecting an emphasis on simplicity and intuitiveness. The third factor included the following two indicators: modern and environmentally friendly, which was named the ecomodern factor. This factor

highlights innovative and ecologically friendly car styling that is consistent with elderly users' expectations for green and innovative NEVs.

Table 2. Kansei semantic evaluation matrix data.

No.	Brief	Bright	Harmonious	Stable	Affinity	Clear	Distinctiveness	Modern	Environmentalism
1	4.50	2.95	3.80	3.70	3.60	3.80	3.50	3.85	3.75
2	3.90	3.90	4.15	4.05	3.80	3.10	3.80	3.50	3.65
3	4.10	4.55	4.20	4.10	3.90	3.90	3.85	4.15	3.80
4	4.25	3.40	3.85	3.70	3.75	3.70	3.25	3.80	3.95
5	4.05	3.45	4.15	4.00	3.60	3.45	3.85	3.75	3.95
6	4.05	3.75	3.95	3.70	3.95	4.05	3.75	3.90	3.55
7	3.75	3.95	3.90	3.95	3.65	3.30	3.65	3.55	3.60
8	4.20	4.65	4.40	4.30	4.15	3.85	4.15	3.85	4.00
9	4.15	3.55	3.80	3.65	3.70	3.60	3.75	3.80	3.70
10	3.95	3.90	3.95	3.90	3.80	3.90	3.70	3.75	4.00
11	4.10	4.15	4.00	3.85	3.90	3.80	3.80	3.55	3.65
12	3.95	3.90	4.10	3.55	3.70	3.50	3.60	3.55	3.60
13	4.30	3.00	3.85	4.20	3.60	3.70	3.30	3.70	3.90
14	3.95	3.20	4.00	3.80	3.35	3.80	3.20	3.20	3.40
15	4.05	3.90	3.90	4.10	3.75	3.90	3.90	3.85	3.85
16	3.90	3.95	3.85	4.10	3.85	3.65	3.85	3.70	3.60
17	4.00	3.95	4.05	3.75	3.55	3.85	3.65	3.45	3.40
18	4.00	4.20	3.95	4.00	4.00	3.55	3.90	3.90	3.80
19	4.00	4.35	3.50	3.90	3.60	3.50	3.75	3.40	3.55
20	4.10	4.15	4.10	4.00	3.85	4.05	3.80	3.85	3.70
21	3.65	2.80	3.05	3.35	2.95	3.30	3.20	2.95	3.20
22	3.60	3.05	3.50	3.60	3.20	3.15	3.35	2.95	3.25
23	4.10	2.95	3.05	3.20	3.60	3.65	3.15	3.35	3.40
24	4.30	3.95	3.90	3.50	3.90	3.75	3.60	3.90	3.70
25	3.75	2.90	3.45	3.50	3.55	3.40	3.50	3.35	3.55
26	3.90	3.05	3.55	3.30	3.35	3.55	3.40	3.45	3.90
27	3.75	3.75	4.05	4.05	4.00	3.60	3.95	4.00	3.65
28	3.75	3.75	3.50	3.60	3.60	3.30	3.45	3.55	3.65
29	3.65	3.40	3.70	3.50	3.40	3.00	3.65	3.20	3.50
30	4.05	3.60	4.10	3.90	4.00	3.70	3.85	4.00	4.05

Table 3. Factor analysis test results.

Kaiser–Meyer–Olkin Measure of Sampling Adequacy		**0.861**
Bartlett's Test of Sphericity	Approx. chi-square	189.135
	df	36
	Sig.	0.000

Table 4. Factor analysis results for the Kansei words.

IE	Sum of Squared Loadings						Sum of Squared Rotated Loadings		
	Sum	Var./%	Cum./%	Sum	Var./%	Cum./%	Sum	Var./%	Cum./%
1	5.36	59.551	59.551	5.36	59.551	59.551	3.701	41.121	41.121
2	1.503	16.7	76.251	1.503	16.7	76.251	2.115	23.503	64.623
3	0.665	7.386	83.637	0.665	7.386	83.637	1.711	19.014	83.637
4	0.493	5.476	89.114						
5	0.274	3.047	92.16						
6	0.246	2.738	94.899						
7	0.209	2.327	97.226						
8	0.157	1.739	98.965						
9	0.093	1.035	100						

Table 5. Factor loading matrix.

Kansei Word	Component		
	1	2	3
Brief		0.816	
Bright	0.911		
Harmonious	0.745		
Stable	0.75		
Affinity	0.725		
Clear		0.913	
Distinctiveness	0.908		
Modern	0.521	0.534	0.555
Environmentalism			0.881

Table 6. Factor score coefficient matrix.

Kansei Word	Component		
	1	2	3
Brief	−0.204	0.436	0.131
Bright	0.39	0.093	−0.435
Harmonious	0.179	−0.057	0.103
Stable	0.2	−0.163	0.145
Affinity	0.162	0.103	−0.003
Clear	0.012	0.662	−0.434
Distinctiveness	0.318	−0.159	−0.045
Modern	0.003	0.119	0.241
Environmentalism	−0.179	−0.224	0.823

3.3. RST: Identification of the Key Design Features of Low-Speed NEVs

The morphological decomposition of the samples in the database (Figure 5) was carried out, and the RST's attribute reduction algorithm was applied to explore the relationships between the visual design features of low-speed NEVs and the degree of satisfaction with the elderly-oriented shape design. The key design characteristics that significantly influence the elderly-oriented degree of satisfaction were extracted. Firstly, a morphological analysis was conducted to decompose the shapes of 100 low-speed NEVs in the sample database into eight design features, which were "Hub", "Side window", "Intake grille", "Headlight", "Car side view", "Rearview mirror", "Front door", and "Fog light". Each design feature was divided into 80 design categories, as depicted in Figure 6. Subsequently, 100 low-speed NEV samples from the database were linked to the design characteristics in the morphological analysis table, as shown in Table 7.

Design Features	Type1	Type2	Type3	Type4	Type5	Type6	Type7	Type8	Type9	Type10
Hub										
Side window										
Intake grille										
Headlight										
Car side view										
Rearview mirror										
Front door										
Fog light										

Figure 6. From the deconstruction table.

Table 7. Sample form: feature deconstruction matrix.

No.	Hub	Side Window	Intake Grille	Headlight	Car Side View	Rearview Mirror	Front Door	Fog Light
1	**4**	**3**	**2**	**1**	**1**	**4**	**9**	**10**
2	9	7	2	2	1	2	9	4
3	2	7	10	2	1	2	5	4
...	...	...	...	...	...	...	...	...
98	6	3	7	9	2	10	1	7
99	4	10	10	4	10	3	5	8
100	2	8	3	10	10	4	5	7

Each design feature of the forms of the low-speed NEVs had different levels of importance in relation to elderly-oriented satisfaction with the overall car form, so the design features that have no or little influence on elderly-oriented satisfaction concerning the low-speed NEVs' forms may lead to inaccurate conclusions with the subsequently used method. Therefore, the RST attribute reduction algorithm was applied in this study to reduce the dimensions of the design feature data, thus obtaining higher precision recognition. Firstly, a focus group (10 people aged 45 to 60 and 10 car designers) discussed the evaluation of an elderly-oriented degree of 80 design features after the evaluation segmentation, as shown in Table 8. In view of the fact that the elderly often find it difficult to maintain concentrated attention because of reduced of physiological and psychological functions, the elderly-oriented degree was defined as follows: a lower elderly-oriented satisfaction degree was labeled as a discrete value of 1, while a higher elderly-oriented satisfaction degree was labeled as discrete value of 3 to provide a clear selection basis for the elderly. Table 8 shows the elderly-oriented satisfaction evaluation matrix with design features. Secondly,

100 subjects (50 males and 50 females aged 45–65 years old, with over 15 years of driving experience, from Nanjing, Jiangsu Province, China) were chosen to evaluate the elderly-oriented satisfaction with the 100 low-speed NEVs in the sample database. The evaluation method adopted a five-order Likert scale. The specific tasks and environment of the experiment were the same as those for the perceptual evaluation. The findings are shown in Table 9. The data in Table 9 were imported into IBM SPSS Statistics 27 software for visual sub-box processing, and the elderly-oriented degree was set as follows: the lower elderly-oriented satisfaction was calibrated as a discrete value of 1; while the higher elderly-oriented satisfaction degree was calibrated as a discrete value of 3. The condition attributes of RST were set as Hub, Side window, Intake grille, Headlight, Car side view, Rearview mirror, Front door, and Fog light, which was set as $C = \{c_1, c_2, c_3, c_4, c_5, c_6, c_7, c_8\}$. The elderly-oriented satisfaction degree of the low-speed NEV forms was set as decision attribute D. The formed RST discretization decision table is shown in Table 10.

Table 8. Design features of low-speed NEV and elderly-oriented satisfaction evaluation matrix.

Design Feature	1	2	3	4	5	6	7	8	9	10
Hub	1	2	3	2	2	3	3	3	2	1
Side window	1	1	2	2	1	2	3	3	3	3
Intake grille	3	3	2	2	2	1	3	2	2	3
Headlight	2	3	2	1	1	1	3	2	3	2
Car side view	3	2	1	3	3	2	1	2	2	3
Rearview mirror	1	3	2	2	3	2	1	2	3	2
Front door	3	1	2	2	2	3	1	2	3	1
Fog light	2	2	3	3	1	1	3	3	1	1

Table 9. Low-speed NEV sample elderly-oriented satisfaction evaluation matrix.

No.	Pts	No.	Pts	No.	Pts	No.	Pts	No.	Pts
1	3.45	21	3.20	41	3.20	61	3.65	81	3.55
2	3.35	22	3.30	42	3.25	62	3.25	82	3.50
3	3.25	23	3.25	43	3.40	63	3.05	83	3.15
…	…	…	…	…	…	…	…	…	…
18	3.95	38	3.75	58	3.35	78	3.45	98	3.20
19	3.30	39	3.15	59	3.20	79	3.55	99	3.05
20	3.50	40	3.25	60	3.60	80	3.35	100	3.25

Table 10. Low-speed NEV elderly-oriented discrete decision table.

U	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8	D
1	2	2	3	2	3	2	3	1	2
2	2	3	3	3	3	3	3	3	2
3	2	3	3	3	3	3	2	3	1
…	…	…	…	…	…	…	…	…	…
98	3	2	3	3	2	2	3	3	1
99	2	3	3	1	3	2	2	3	1
100	2	3	2	2	3	2	2	3	1

MATLAB R2022b software was utilized for the RST attribute reduction algorithm to pinpoint key design characteristics that have a significant impact on elderly-oriented satisfaction. The outcomes are presented below:

Initially, the table data were separated into equivalence classes according to the conditional attributes and decision attributes, as follows:

$U/IND(C) = \{(1),(2),(3),(4),(5),(6),(7),(8),(9),(10),(11),(12),(13),(14),(15),(16),(17),\ (18),\ (19),\ (20),(21),(22),$
$(23),(24),(25),(26),\ (27),(28,65),(29),(30),\ (31),(32),(33),(34),(35),(36),\ (37),(38),(39),(40),(41),\ (42),(43),(44),$
$(45),(46),(47),(48),(49),(50),(51),(52),(53),(54),\ (55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66),(67),$
$(68),(69),(70),(71),(72,96),(73),(74),\ (75),(76),(77),(78),(79,93),(80),(81),(82),(83),(84),(85),(86),(87),(88),$
$(89),(90),(91),(92),(94),(95),(97),(98),(99),(100)\}$
$U/IND(D) = \{(1,2,6,7,9,10,11,12,14,15,19,22,26,28,33,37,43,44,46,55,58,66,72,77,78,80,84,85,90,92,97),$
$(3,4,13,17,21,23,24,27,30,31,32,34,36,39,40,41,42,48,50,53,59,62,63,64,69,70,73,74,76,83,86,87,88,93,94,95,96,98,99,100),$
$(5,8,16,18,20,25,29,35,38,45,47,49,51,52,54,56,57,60,61,65,67,68,71,75,79,81,82,89,91)\}$

After individually removing the conditional attributes, the fields were divided into equivalence classes as follows:

$U/IND(C-\{c_1\}) = \{(1),(2),(3),(4),(5,89),(6,20),(7),(8),(9,29),(10),(11),(12),(13),(14),(15),(16),(17,21),(18),(19),$
$(22),(23),(24),(25),(26),(27),(28,65),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),(44),(45),$
$(46),(47),(48),(49),(50),(51,53),(52),(54),(55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66),(67),(68),(69),(70),$
$(71),(72,96)(73),(74),(75),(76),(77),(78),(79,93),(80),(81),(82),(83),(84),(85),(86),(87),(88),(90),(91),(92),(94),(95),$
$(97),(98),(99),(100)\}$
$U/IND(C-\{c_2\}) = \{(1),(2),(3),(4),(5),(6),(7),(8),(9),(10),(11),(12),(13),(14),(15),(16),(17),(18),(19),(20),(21),(22),$
$(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),(44),(45),$
$(46),(47),(48),(49),(50),(51),(52),(53),(54),(55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66),(67),(68),(69),(70),$
$(71),(72,96)(73),(74),(75),(76),(77),(78),(79,93),(80),(81),(82),(83),(84),(85),(86,89),(87),(88),(90),(91),(92),(94),$
$(95),(97),(98),(99),(100)\}$
$U/IND(C-\{c_3\}) = \{(1),(2),(3),(4),(5),(6),(7),(8),(9),(10),(11),(12),(13),(14),(15),(16),(17),(18),(19),(20),(21),(22),$
$(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),(44),(45),$
$(46),(47,74),(48),(49),(50),(51),(52),(53),(54),(55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66),(67,72,96),(68),$
$(69,99),(70),(71),(73),(75),(76),(77),(78),(79,93),(80),(81),(82),(83),(84),(85),(86),(87),(88),(89),(90),(91),(92),(94),$
$(95),(97),(98),(100)\}$
$U/IND(C-\{c_4\}) = \{(1,53),(2),(3,66),(4),(5),(6),(7),(8,20),(9,13),(10),(11),(12),(14),(15),(16),(17),(18),(19),(21),$
$(22),(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),(44),$
$(45),(46),(47),(48),(49),(50),(51),(52),(54),(55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(67),(68),(69),(70),(71),$
$(72,96),(73),(74),(75),(76),(77),(78),(79,93),(80),(81),(82),(83),(84,85),(86),(87),(88),(89),(90),(91),(92),(94),(95),$
$(97),(98),(99),(100)\}$

$U/IND(C-\{c_5\}) = \{(1),(2,94),(3),(4),(5),(6),(7),(8,61),(9),(10),(11),(12),(13),(14),(15),(16),(17),(18),(19),(20),$
$(21),(22),(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),$
$(44,92),(45,85),(46),(47),(48),(49),(50),(51),(52),(53),(54),(55),(56),(57),(58),(59),(60),(62),(63),(64),(66),(67),(68),$
$(69),(70),(71),(72,96),(73),(74),(75,82),(76),(77),(78),(79,93),(80),(81),(83),(84),(86),(87),(88),(89),(90),(91),(95),$
$(97),(98),(99),(100)\}$
$U/IND(C-\{c_6\}) = \{(1),(2),(3),(4),(5),(6,84),(7),(8,),(9),(10),(11),(12,88,97),(13),(14),(15),(16),(17),(18),(19),(20),$
$(21),(22),(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),$
$(44),(45),(46),(47),(48),(49),(50),(51),(52),(53),(54),(55,82),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66,99),(67),$
$(68),(69),(70),(71),(72,96),(73),(74,86),(75),(76),(77),(78),(79,93),(80),(81),(83),(85),(87),(89),(90),(91),(92),(94),$
$(95),(98),(100)\}$
$U/IND(C-\{c_6\}) = \{(1),(2),(3),(4),(5),(6,84),(7),(8,),(9),(10),(11),(12,88,97),(13),(14),(15),(16),(17),(18),(19),(20),$
$(21),(22),(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(38),(39),(40),(41),(42),(43),$
$(44),(45),(46),(47),(48),(49),(50),(51),(52),(53),(54),(55,82),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66,99),(67),$
$(68),(69),(70),(71),(72,96),(73),(74,86),(75),(76),(77),(78),(79,93),(80),(81),(83),(85),(87),(89),(90),(91),(92),(94),$
$(95),(98),(100)\}$
$U/IND(C-\{c_7\}) = \{(1),(2,3),(4),(5),(6),(7),(8),(9),(10),(11),(12),(13),(14),(15),(16),(17),(18),(19),(20),(21),(22),$
$(23),(24),(25),(26),(27,38),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(39),(40),(41),(42),(43),(44,85),(45,92),$
$(46),(47),(48),(49),(50),(51),(52),(53),(54),(55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66),(67),(68),(69),(70),$
$(71),(72,90,96),(73),(74),(75),(76),(77),(78),(79,93,99),(80),(81),(83),(84),(86),(87),(88),(89),(91),(94),(95),(97),$
$(98),(100)\}$
$U/IND(C-\{c_8\}) = \{(1),(2),(3),(4),(5),(6),(7),(8),(9),(10),(11),(12),(13),(14),(15),(16),(17),(18,80),(19),(20),(21),$
$(22),(23),(24),(25),(26),(27),(28,65),(29),(30),(31),(32),(33),(34),(35),(36),(37),(39),(40),(41),(42),(43),(44),(45),$
$(46),(47),(48),(49),(50),(51),(52),(53),(54),(55),(56),(57),(58),(59),(60),(61),(62),(63),(64),(66),(67),(68),(69),(70),$
$(71),(72,96),(73),(74),(75),(76),(77),(78),(79,93),(81),(82),(83),(84),(85),(86),(87),(88),(89),(90),(91),(92),(94),(95),$
$(97),(98),(99),(100)\}$

The cardinality of the decision attribute and each condition attribute was calculated as follows:

$$card[pos_c(D)] = 100,\ card\left[pos_{c-|c_1|}(D)\right] = 88,\ card\left[pos_{c-|c_2|}(D)\right] = 92,\ card\left[pos_{c-|c_3|}(D)\right] = 91,\ card\left[pos_{c-|c_4|}(D)\right] = 88,$$
$$card\left[pos_{c-|c_5|}(D)\right] = 90,\ card\left[pos_{c-|c_6|}(D)\right] = 87,\ card\left[pos_{c-|c_7|}(D)\right] = 86,\ card\left[pos_{c-|c_8|}(D)\right] = 92$$

Therefore, the dependence degree of the decision attribute D on the conditional attribute C was $r_c(D)$; the dependence degree of the decision attribute D on knowledge $C - \{c_e\}$ was $r_{c-|c_e|}(D)$. The results are as follows:

$$r_c(D) = 100/100,\ r_{C-c_1}(D) = 88/100,\ r_{C-c_2}(D) = 92/100,\ r_{C-c_3}(D) = 91/100,\ r_{C-c_4}(D) = 88/100,\ r_{C-c_5}(D) = 90/100,$$
$$r_{C-c_6}(D) = 87/100,\ r_{C-c_7}(D) = 86/100,\ r_{C-c_8}(D) = 92/100$$

Using Formula (4), the importance of the conditional attribute to the decision attribute is calculated, and the result is as follows:

$$\sigma(c_1) = 12/100,\ \sigma(c_2) = 8/100,\ \sigma(c_3) = 9/100,\ \sigma(c_4) = 12/100,\sigma(c_5) = 10/100,\ \sigma(c_6) = 13/100,\ \sigma(c_7) = 14/100,\ \sigma(c_8) = 8/100$$

The weight coefficient was calculated according to Formula (5), as follows:

$$w_1 = 0.15789,\ w_2 = 0.05263,\ w_3 = 0.07895,\ w_4 = 0.15789,\ w_5 = 0.10526,\ w_6 = 0.18421,\ w_7 = 0.21053,\ w_8 = 0.05263$$

Therefore, after the reduction by the attribute reduction algorithm of RST, the weight coefficients of various low-speed NEV form features' influence on elderly-oriented satisfaction are shown in Table 11. The expert group comprehensively considered the reduction of the design features with a weight coefficient of the low-speed NEV design features of less than 0.1, namely, Hub, Headlight, Car side view, Rearview mirror, and Front door.

Table 11. Design feature weighting factors.

Hub	0.15789	Side window	0.05263	Intake grille	0.07895	Headlight	0.15789
Car side view	0.10526	Rearview mirror	0.18421	Front door	0.21053	Fog light	0.05263

3.4. SVR Build of the Mapping Model between Kansei Semantics and Key Features of Low-Speed NEVs

A five-order Likert scale was constructed by combining three kinds of elderly-oriented Kansei words factors clustered in the first stage of this study with 100 low-speed NEV samples in the database. A total of 200 subjects participated in the Kansei intention evaluation. The study population, specific tasks, and settings were the same as the Kansei intention analysis for the factor analysis. The matrix between the evaluation value of the elderly-oriented Kansei factors and the key design features of low-speed NEVs was produced, as shown in Table 12.

Table 12. Key features and Kansei evaluation matrix.

No.	Hub	Headlight	Car Side View	Rearview Mirror	Front Door	Welcome	Clarity	Ecomodern
1	4	1	1	4	9	3.40	3.10	3.20
2	9	2	1	2	9	3.25	3.35	3.15
3	2	2	1	2	5	3.45	3.35	3.35
...	...	...	...	...	...	...	...	...
98	6	9	2	10	1	3.55	3.50	3.40
99	4	4	10	3	5	3.55	3.80	3.25
100	2	10	10	4	5	3.25	3.45	3.65

Using the MATLAB R2022b software platform, SVR was applied to program and calculate the matrix data in Table 12, thus obtaining the set point and mapping functions of the optimized parameters. The elderly-oriented Kansei factor of "Welcomeness" was taken as an example; 90 samples are used as the training set and the remaining 10 samples were used as the test set. When the SVR model in this study was constructed, the grid search method was used in combination with five-fold cross-validation technology to train and verify the model. After continuous attempts and evaluations, we found that the optimal parameter penalty coefficient, C, and the width parameter of the Gaussian radial basis (gamma) were (0.70711, 11.3137), and the converge for better MSE and MBE parameters were (0.997943, 1.3970×10^{-4}, and -1.4698×10^{-4}). The results indicate that the model in this study demonstrates strong predictive capabilities, a high level of accuracy, and minimal variance between the predicted and actual values, as illustrated in Figure 7. The test set was employed to validate the model's performance. A comparison between the actual perceived evaluation values from the test set and the predicted perceived evaluation values of the model is depicted in the Figure 7. The parameters of R^2, MSE, and MBE were (0.99623, 7.0492×10^{-2}, and -5.3592×10^{-2}). The result shows that the prediction accuracy of the mapping model constructed in this study was high, which is in line with expectations. Similarly, the optimal penalty coefficient, C, and the width parameter (gamma) of the Gaussian radial basis for the other two elderly-oriented Kansei factors were obtained as (0.70711,32) and (0.5, 22.6274). The parameter results for the test set R^2, MSE, and MBE were (0.998835, 9.8112×10^{-5}, -2.8806×10^{-4}), (0.997911, 1.59023×10^{-4}, and -1.3444×10^{-4}); the parameter results of the training set, MSE, and MBE were predicted as (0.99725, 2.76051×10^{-2}, and -4.5223×10^{-2}) and (0.99603, 1.73261×10^{-2}, and 3.368×10^{-3}), which are shown in Figure 7 and Table 13. The findings of this study show that the SVR mapping model constructed in this study has high prediction accuracy and can be used to establish the mapping relationship between users' emotional feelings and the design features of low-speed NEVs.

Ultimately, to identify the low-speed NEV design features that provide the greatest elderly-oriented satisfaction that correspond to the design characteristics with the highest average of the elderly-oriented Kansei factors, five key design features were determined through rough set reduction. Each design feature was associated with 10 design categories in this stage of the study. Therefore, there were a total of $10 \times 10 \times 10 \times 10 \times 10 = 100{,}000$ design combinations. All design combinations were computer coded and used as input parameters in the SVR model to predict the evaluated value of the corresponding three elderly-oriented Kansei factors for each combination, as shown in Table 14. For each design combination, the maximum average values of the three elderly-oriented Kansei factors served as criteria for assessing elderly-oriented satisfaction. The experimental findings reveal that the highest mean perception evaluation value was 3.523282, indicating that the optimal design feature combination for the elderly-oriented design of low-speed NEV body shapes was identified, and the corresponding design feature categories were hub type six, headlight type nine, car side view type two, rearview mirror type nine, and front door type ten. Using the method of taking the average maximum value of each factor, the assessment value for each elderly-oriented Kansei factor was taken into account in a balanced way. It is believed that the product form design with high elderly-oriented satisfaction in all aspects will be more in line with the Kansei intentions of elderly users.

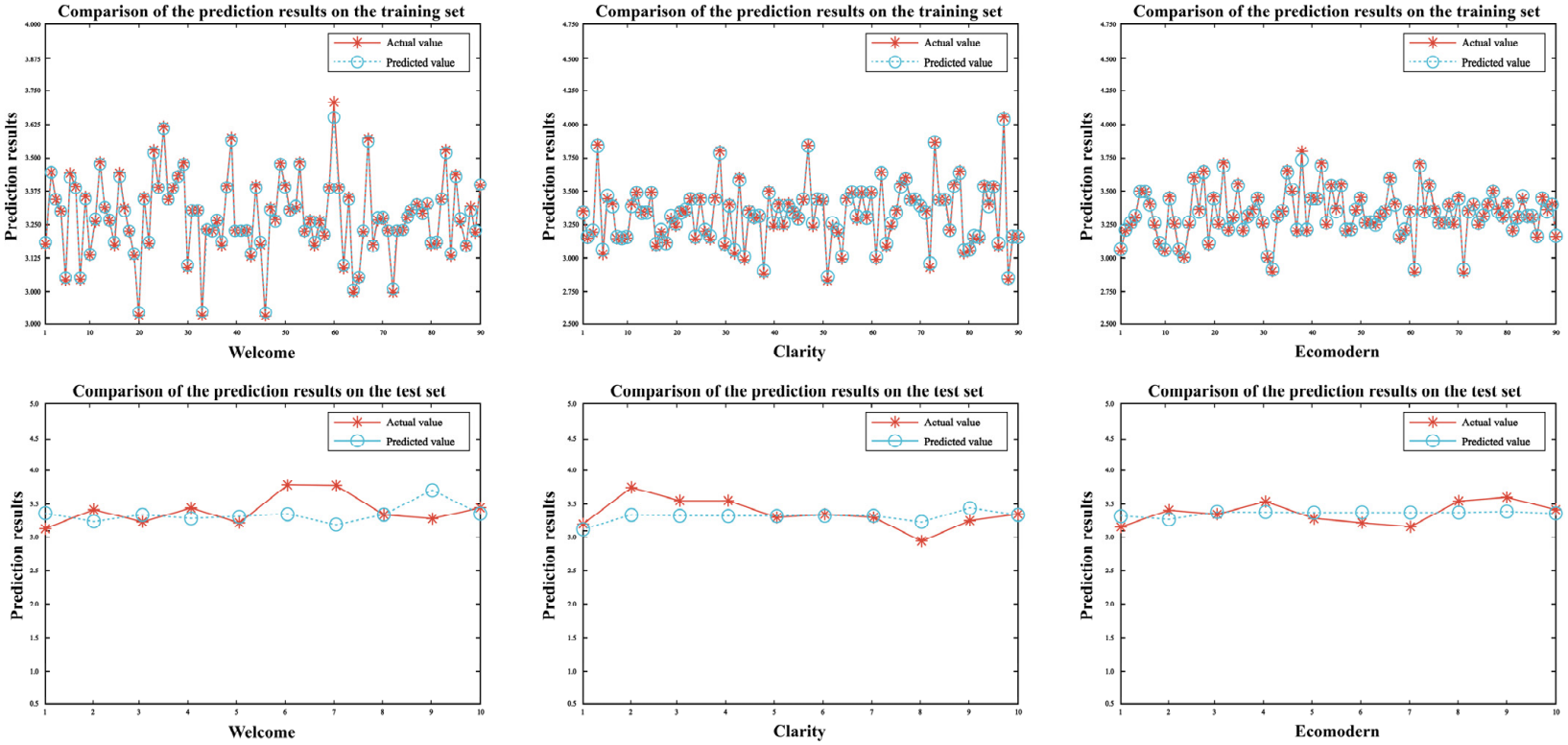


Figure 7. Fitted plots of the training and test sets.

Table 13. The parameter results matrix.

Kansei Word	Hyperparameters			Results on the Training Set			Results on the Test Set	
	C	Gamma	R^2	MSE	MBE	R^2	MSE	MBE
Welcome	0.70711	11.3137	0.997943	1.3970×10^{-4}	-1.4698×10^{-4}	0.99623	7.0492×10^{-2}	-5.3592×10^{-2}
Clarity	0.70711	32	0.998835	9.8112×10^{-5}	-2.8806×10^{-4}	0.99725	2.76051×10^{-2}	-4.5223×10^{-2}
Ecomodern	0.5	22.6274	0.997911	1.59023×10^{-4}	-1.3444×10^{-4}	0.99603	1.73261×10^{-2}	3.368×10^{-3}

Table 14. Mapping model predictive value matrix.

No.	Welcome	Clarity	Ecomodern	Average
1	3.314446	3.326493	3.321654	3.320864
2	3.314595	3.326116	3.322058	3.320923
3	3.314595	3.326116	3.322058	3.320923
...	...	...	...	...
58,191	3.540985	3.487989	3.540870	3.523282
...	...	...	...	...
9998	3.308972	3.326236	3.321663	3.318957
9999	3.308972	3.326236	3.321663	3.318957
100,000	3.307404	3.326236	3.321654	3.318431

For the purpose of verifying the effectiveness of the proposed method, we took the experimental results as a reference, namely, hub type six, headlight type nine, car side view type two, rearview mirror type nine, and front door type ten were selected in the morphological deconstruction table. For the other design characteristics, the design classifications with high elderly-oriented satisfaction in Table 8 are selected. Combined with the main aesthetic style and creativity of the designers, Adobe Photoshop 2022 software was used for the sketches. Zbrush 2021.7 software was used to transform the 2D sketches into 3D relief effects, and then Stable Diffusion software was used to adjust the design parameters to generate high-quality image prediction (Figure 8) and render 3D models so that the design of an elderly-oriented form of a low-speed NEV was obtained, as shown in Figure 9. The low-speed NEV featured in Figure 9 was evaluated by 200 elderly participants using a five-point Likert scale, resulting in an average score of 4.05. This score was higher than the value obtained from 100 samples, shown in Table 9, which strongly validates the effectiveness and practicality of the proposed combination method for elderly-oriented form design. Product designers can utilize this method as a reference to inform the design of products targeted towards the elderly, ensuring that they cater to the emotional needs of elderly users.



Figure 8. Product manufacturing process.



Figure 9. Elderly-oriented design of a low-speed NEV.

4. Analysis and Discussion of Results

4.1. Analysis of Results

As the aging global population problem becomes increasingly serious, elderly-oriented design has become an important issue. Elderly-oriented designs not only need to meet the needs of elderly users but also adjust the design parameters to satisfy their emotional needs. In the process of elderly-oriented design, it is necessary to fully take the physiological characteristics and cognitive ability of elderly users into consideration, thus ensuring the integrity and applicability of low-speed NEV body's form. Previous studies generally relied on qualitative analysis to evaluate elderly-oriented design, but this lacks qualitative and quantitative modeling methods. At present, there is no systematic research on the elderly-oriented form design of low-speed NEVs. In view of this, a combined method that took KE as the research framework and RST and SVR to predict the products' elderly-oriented form design was used. The low-speed NEV was taken as an example to verify the effectiveness of the proposed method. Firstly, the low-speed NEV with a 45-degree squint was selected as the study object; three elderly-oriented Kansei words factors were obtained by FA dimensionality reduction clustering, namely, Welcome, Clarity, and Ecomodern. Through morphological analysis, eight design features of low-speed NEVs were obtained and further subdivided into 80 design types. Because of the wide variety of design features, this could easily affect the computational efficiency of the subsequent algorithm, and the design characteristics that did not have an influence on elderly-oriented satisfaction would cause the subsequently calculated results to be inaccurate as well. Therefore, the attribute reduction algorithm of RST was applied to calculate the weights of the design features, thus determining the priority of each design feature's influence on elderly-oriented satisfaction. After reducing three kinds of design features, five key design features were obtained, namely, Hub, Headlight, Car side view, Rearview mirror, and Front door. Finally, the morphological deconstruction of the key design characteristics and the evaluated value of the elderly-oriented Kansei words of the 100 samples were introduced into the SVR training model. The experimental results show that the R^2 of the three mapping functions were all greater than 0.95, indicating a good fitting effect. The model was used to predict the combination of design features with the highest average value of elderly-oriented Kansei factors so that the optimal combination of elderly-oriented form design features for low-speed NEVs was obtained. The results show that from the form analysis table, hub type six, headlight type nine, car side view type two, rearview mirror type nine, and front door type ten can satisfy the Kansei needs of elderly users to the maximum extent.

In this study, RST and SVR were applied to build a prediction model for Kansei intention and shape design features of products for elderly users, which can successfully predict the combination of elderly-oriented design features for products' shapes in the future and transform the subjective process of selection into a scientific and objective selection process. Automobile designers can use this method as a basis to select elderly users in cities, record their emotional needs, identify key morphological features, and draw sketches. Then, they can use artificial intelligence technology to generate images and develop elderly-oriented and low-speed NEV designs for specific cities, thus better meeting the Kansei requirements of elderly users. In view of this, the method can serve as a direct reference for enterprises, helping them expand their elderly consumer base, improve the quality of design decisions, shorten the product development cycle, and significantly increase product sales and the market value of enterprises.

4.2. Discussion

In KE, determining the weights of design feature indicators and building a mapping model between the emotional preferences of users and the design features of products are the core steps. Previous methods to measure the index weights and calculate the design feature index weights, such as the QFD and KANO models, have difficulty in handling large-scale data, and they ignore the impact of users' emotions on product experience. Therefore, the RST attribute reduction algorithm of artificial intelligence technology was chosen for use in this study instead of traditional methods to quickly and accurately calculate the key design characteristics that significantly influence elderly-oriented satisfaction. Meanwhile, when the mapping model was constructed, the previous linear statistical method of principal component analysis (PCA) and mixed linear discriminant analysis (MLDA) were used. They are unable to deal with high-latitude and highly correlated nonlinear data and cannot capture the complex relationship and nonlinear dynamic changes in Kansei emotions directly. With the development of artificial intelligence technology, neural network and deep learning methods have been successfully applied to measure users' fuzzy Kansei knowledge. Hu et al. [64] combine a convolutional neural network (CNN) to construct a users' perception evaluation system. Li et al. [65] combined RST and a backpropagation neural network (BPNN), as well as other methods, to create and verify a KE evaluation system and predict key morphological characteristics of products based on a certain KE word. Kang et al. [66] combine KE and the interactive genetic algorithm (IGA) to build a product form design system and optimize the decision-making process of cultural and creative product designs. Lian et al. [67] combine KE with GST and SVR to establish an accurate prediction model of product images, with which designers can correctly evaluate the blue emotional product shape design. Lin et al. [68] combine KE with the quantification theory type (QTTI), BPNN, and a genetic-algorithm-based BPNN, among other methods. The model constructed by BPNN proved to be the most accurate and provided an accurate sound design index and reference. Based on the KE engineering system, consumption was modeled and the product morphological characteristics were analyzed by Yang et al. [69] to determine the key morphological characteristics. In summary, most of the users in the above studies were young, while elderly users were ignored. Consequently, the design solutions cannot meet the emotional needs of all social groups. The contrast between this study and previous studies is shown in Table 15. At the same time, artificial intelligence technology was used in all these studies to build mapping models. After the discussion, the expert group made comparisons of the artificial technology used in perception mapping, as shown in Table 16. From the comprehensive analysis, SVR performed well in dealing with small samples, nonlinear problems, error control, and pattern recognition for high-dimensional data. It could also effectively avoid problems related to the local minimum and overlearning, which makes it suitable for building Kansei mapping models for elderly users.

Table 15. Comparison of the present study with previous studies.

Publication	Methods	User	Construct Mapping Models	Research Field
This paper	SVR	elderly user	Kansei evaluation and product form features	Product design (low-speed NEV)
Hu et al. [64]	CNN	young user	Kansei evaluation and product form features	Product design (weight scales and goblets)
Li et al. [65]	BPNN	young user	Kansei evaluation and product key form features	Product design (footwear)
Kang et al. [66]	IGA	young user	Kansei evaluation and cultural symbols	Product design (creative industries)
Lian et al. [67]	SVR	young user	Kansei evaluation and product form features	Product design (numerical control machine)
Lin et al. [68]	QTTI, BPNN	young user	Kansei evaluation and sound elements	Product design (electric shaver)
Yang et al. [69]	SVM	young user	Product performance and product form features	Product design (digital camera)

Table 16. Contrasting the different sentimental mapping algorithms.

Algorithm	Algorithm Type	Scope of Application	Data Requirements	Adjustable Parameters	Training Speed	Prediction Speed	Memory Consumption
BPNN	Neural Network	Complex pattern recognition, nonlinear relationships	Bulk labeled data, normalized inputs	Number of layers, number of nodes, learning rate	Slow	Fast	High
GST	Ensemble Method	Regression and classification issues	Strong ability to handle missing values, numerical type characterization	Tree depth, learning rate, subsampling	Medium	Fast	Medium
SVR	Support Vector Machine	Regression on issues	Linearly separable or approved separable data	Kernel function types, regularized arguments	Medium	Medium	Low
IGA	Evolutionary Algorithm	Optimization Issues	No need for gradient information, multimodal problems	Population size, crossover rate, mutation rate	Medium	Low	Low
PCA	Dimensionality Reduction	Feature extraction, data compression	Numerical data, linear relationships	Principal component, explained variance ratio	Fast	Fast	Low
QT-I	Metaheuristic Algorithm	Regression on issues	No need for gradient information, multimodal problems	Taboo length, number of iterations, neighborhood search	Slow	Slow	Low

Therefore, combined with low-speed NEVs, there were multidimensional graphical data. In this study, a combination method for the elderly-oriented design of product shape is proposed based on KE/RST/SVR. In a comparison with the previous KE, the artificial intelligence technologies of RST and SVR are combined in this study to assist in design decision making, and they could process nonlinear Kansei data of elderly users, extract the key design features using the RST attribute reduction algorithm, construct nonlinear mapping functions between design features and elderly users' Kansei evaluations by SVR, and predict the optimal combination of design features. The findings of this study provide an effective reference for designers in the early stage of low-speed elderly-oriented form design and give full play to designers' creativity while satisfying the emotional requirements of elderly users. In summary, the elderly-oriented design combination method

solves the problems of large number of calculations, lack of linear model fitting, and lack of user groups in traditional KE research. Combined with KE and artificial intelligence technology, an elderly-oriented design method for product forms is proposed in this paper, which can design a product form favored by elderly users for enterprises to use in combination with designers' creativity.

4.3. Impact of the Design of Elderly-Oriented NEVs on Market Value and Sustainability

Based on engineering theory and in combination with elderly users' sentiment-based feedback, a low-speed NEV was designed in this paper. This research is dedicated to the development and design of elderly-oriented low-speed NEVs that take the needs and differences of the elderly population into consideration, increases inclusion in society, and provides new insights into the market value of enterprise and the sustainable development of society. As the global aging trend intensifies, enterprises are paying more attention to elderly-oriented design processes [70]. By optimizing product design [71] and improving public measures and service modes [72], elderly-oriented design enables elderly users to better participate in social life, enjoy social welfare, and improve the quality of life, thus promoting the long-term and sustainable development of society. The popularization of elderly-oriented design can not only help elderly people participate in social activities more independently but also reduce their gap with the young people. Therefore, this study will have a positive impact on the sustainable development of society. From a broader social perspective, NEV design targeting the elderly not only promotes the inclusion of the elderly but also promotes environmental protection and climate responsibility. These vehicles reduce dependence on fossil fuels through the use of new energy technologies, which is in line with the goal of sustainable development. In addition, by designing new energy vehicles that can attract the elderly, technological innovations and the business model process of this market segment can also be promoted [73]. According to automobile sales' market statistics, the scale and proportion of middle-aged and elderly consumers 45 years of age or older continues to increase. At the same time, with the passage of time, growth in the group of car owners aged 35 to 45 years old over the next 10 years will increase significantly [74]. Therefore, automobile enterprises need to carry out elderly-oriented product design for elderly consumer groups so as to cater to this huge and growing market demand, expand elderly consumer groups, and increase market shares. Policymakers and industry regulators can encourage research development subsidies and financial support by developing dedicated vehicle design standards and guidelines. They can also carry out public education and propaganda activities, as well as cooperate with industry associations, universities and research institutions to support the development and application of vehicle designs for elderly people. At the same time, testing and feedback mechanisms should be established; barrier-free infrastructure should be promoted; relevant laws and regulations should be improved; smart technology should be introduced and special working groups should be created as well; the cross- departmental cooperation should be promoted to ensure that the elderly can enjoy greater convenience and safety while traveling. These measures will be beneficial for driving innovation and the application of vehicle technologies adapted to the needs of the elderly. So this paper is closely related to market value. Elderly consumers focus on the functionality, quality, safety, and comfort of products. Designers need to integrate Kansei aesthetics and elderly-oriented design projects and launch elderly-oriented products to improve the market competitiveness, thus meeting the needs and preferences of elderly users. In order to promote the sustainable development of society and against a background of global population aging, a numerical evaluation model for elderly-oriented low-speed NEVs is proposed in this thesis, hoping to achieve the optimal design of elderly-oriented low-speed NEVs that meet the Kansei needs of elderly users. Low-speed NEVs that are elderly-oriented offer a new solution to the travel difficulties experienced by elderly people, thereby improving the quality of elderly people's lives and enabling them to participate more actively in social activities, effectively reducing the care burden on the younger generation. The research results provide a new concept for

aging societies and promote the sustainable and green development of society. At the same time, the research results can help designers with guidance on parameters during the early stages of developing elderly-oriented products, thus improving the satisfaction of elderly users and product sales, as well as effectively reducing the risks associated with research and development.

5. Conclusions

In this study, KE was taken as the framework. The RST and SVR methods in combination with artificial intelligence technology was used to develop a combination elderly-oriented design method for product form. Low-speed NEVs were taken as the study case to verify the feasibility of the combination method. In the KE research framework, FA clustering was used in the first stage to identify three elderly-oriented Kansei words factors. In the second stage, morphological analysis was used to deconstruct low-speed NEVs. The RST attribute reduction algorithm was used to screen the key morphological design features that have significant influence on elderly-oriented satisfaction. In the third stage, SVR was adopted to construct the mapping function between the Kansei intention of elderly users and the key design features; thus, the optimal combination of a miniature NEV shape design was obtained in line with the emotional preferences of elderly users. In the last stage, we used traditional engineering software and artificial intelligence software to complete the design practice and verify the feasibility of the combined method by evaluating the satisfaction of elderly users. The main contributions of this thesis include the following:

1. Using the research framework of KE, a combination method of KE, RST, and SVR for product form design is proposed.
2. RST was used to identify the key design features of low-speed NEVs that have an important influence on elderly-oriented satisfaction.
3. SVR was used to build a mapping model between elderly-oriented Kansei factors and product form design features in a nonlinear manner.

There are several limitations to this study. First, the subjects of the Kansei intention analysis were limited to elderly users in Nanjing, which may affect the universality and generalization of the research results. Elderly users in different regions may have different cultures, economies, and preferences, so conclusions about the form design of low-speed new energy vehicles may not apply to elderly users in other regions. In addition, KE theory is limited to the use of simple adjectives to describe the complex emotions of elderly users, which cannot fully consider the emotional cognition and emotional response of subjects. Secondly, only one case was used in this study to verify the proposed method, and more cases are needed to prove the generalization of the method. Finally, the research was carried out only from a single perspective using two-dimensional graphics of the product, lacking in-depth research on the interior design of NEVs, the functions, and man–machine elements. It is suggested that future research can be carried out from the following aspects: (1) Expand the sample scope to include more elderly users in more areas to improve the universality of the research results. (2) A physiological signal testing instrument should be used to study elderly users' cognition and perception process for product form design. (3) Methods such as computer-aided design and ergonomics simulation should be used to study the design and function of product interiors in three-dimensional space so as to deeply explore the relationship between new energy vehicles and elderly-oriented requirements. We hope that this study can serve as a practical reference and guidance for the elderly-oriented design of new energy vehicles.

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